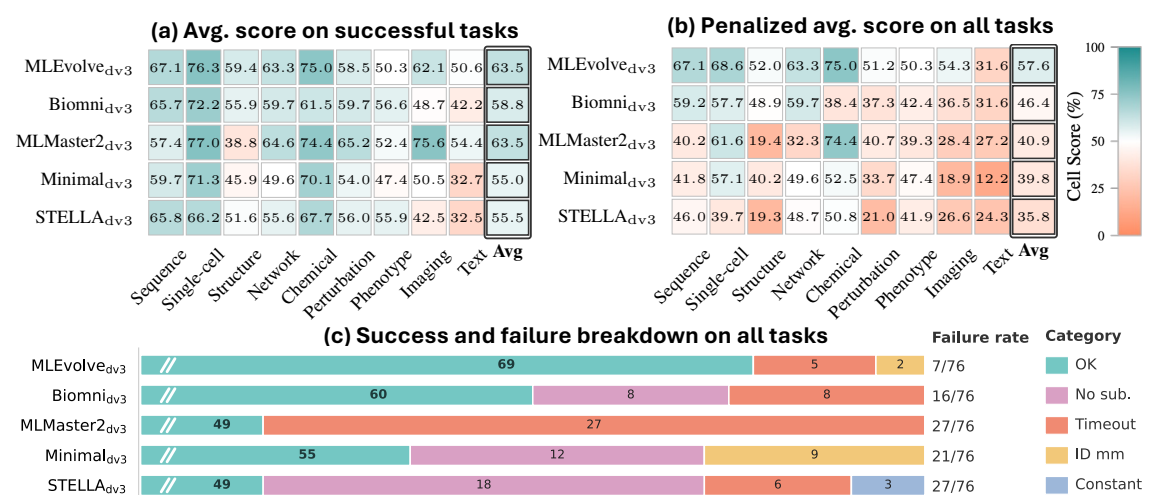

(a) Avg. score on successful tasks
MLEvolve$_{dv3}$ 67.1 76.3 59.4 63.3 75.0 58.5 50.3 62.1 50.6 63.5
Biomni$_{dv3}$ 65.7 72.2 55.9 59.7 61.5 59.7 56.6 48.7 42.2 58.8
MLMaster2$_{dv3}$ 57.4 77.0 38.8 64.6 74.4 65.2 52.4 75.6 54.4 63.5
Minimal$_{dv3}$ 59.7 71.3 45.9 49.6 70.1 54.0 47.4 50.5 32.7 55.0
STELLA$_{dv3}$ 65.8 66.2 51.6 55.6 67.7 56.0 55.9 42.5 32.5 55.5
Sequence
Single-cell
Structure
Network
Chemical
Perturbation
Phenotype
Imaging
Text
Avg
(b) Penalized avg. score on all tasks
MLEvolve$_{dv3}$ 67.1 68.6 52.0 63.3 75.0 51.2 50.3 54.3 31.6 57.6
Biomni$_{dv3}$ 59.2 57.7 48.9 59.7 38.4 37.3 42.4 36.5 31.6 46.4
MLMaster2$_{dv3}$ 40.2 61.6 19.4 32.3 74.4 40.7 39.3 28.4 27.2 40.9
Minimal$_{dv3}$ 41.8 57.1 40.2 49.6 52.5 33.7 47.4 18.9 12.2 39.8
STELLA$_{dv3}$ 46.0 39.7 19.3 48.7 50.8 21.0 41.9 26.6 24.3 35.8
Sequence
Single-cell
Structure
Network
Chemical
Perturbation
Phenotype
Imaging
Text
Avg
Cell Score (%)
100
75
50
25
0
(c) Success and failure breakdown on all tasks
Failure rate
Category
MLEvolve$_{dv3}$ 69 5 2 7/76
Biomni$_{dv3}$ 60 8 8 16/76
MLMaster2$_{dv3}$ 49 27 27/76
Minimal$_{dv3}$ 55 12 9 21/76
STELLA$_{dv3}$ 49 18 6 3 27/76
OK
No sub.
Timeout
ID mm
Constant